Online implemented document store

Connection string:
mongodb+srv://st126128_db_user:zIfMACQ2Bmp8oRHi@cluster0.bbjbnzd.mongodb.net/
Database name: CSSS

Project description and scope definition

Usecase: Citizen Sourcing-based System for Smart City and Smart Governance of Pathumthani (CSSS).
Our document model has two documents: **Issues** and **Solutions**.

Solutions:

Each solution describes the actions taken to fix one or more related issues.
Attributes include:

- **solution_id:** unique identifier for the solution,
- **issues**: list of related issue IDs - an issue reference,
- **details**: description of the fixing plan or actions,
- **office**: the responsible government office (name, contact, address),
- **start_date** and **finish_date**: solution project timeline,
- **cost**: estimated or actual cost of the fix,
- **feedback**: citizen feedback after completion (rating, review, citizen reference)


```
{
  "solution_id": 1
  "issues": [{"issue_id": 1}, {"issue_id": 2}],
  "details": "Repair and expand the drainage system.",
  "cost": 75932,
  "office": {"name": "Public Works Department", "contact": "02-123-4567"},
  "start_date": "2025-03-12",
  "finish_date": "2025-04-10",
  "feedback": [
    {
      "rating": 5,
      "review": "Good response, fixed quickly.",
      "citizen": {"first_name": "Anna", "last_name": "Smith", "thai_id": "3720465060935" }
    },
    {
      "rating": 3,
      "review": "The issue was not fixed quickly",
      "citizen": {"first_name": "Bob", "last_name": "Jones", "thai_id": "3749829060543" }
```

```
    }
  ]
}
```

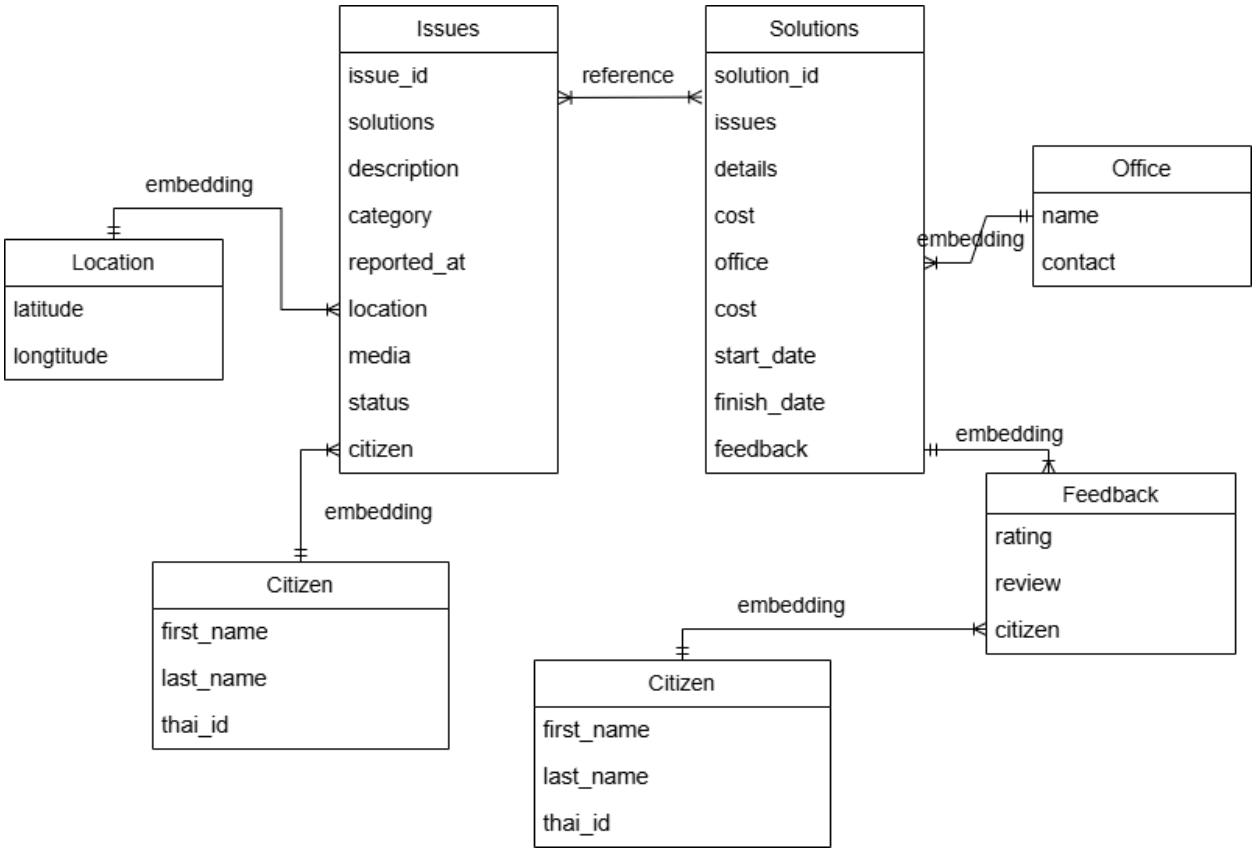
Issues:

Each issue represents a problem reported by a single citizen.
Attributes include:

- **issue_id**: unique identifier for the issue,
- **solutions**: list of related solution IDs - a solution reference,
- **description**: details of the reported problem,
- **category**: classification (e.g., flood, waste, lighting),
- **reported_at**: date and time of reporting,
- **location**: geographical coordinates (longitude, latitude),
- **media**: attached evidence (photos, videos),
- **status**: can be pending, in progress, or resolved,
- **citizen**: information about the reporter (name, ID)

```
{
  "issue_id": 1,
  "solutions": [{"solution_id": 5}, {"solution_id": 7}],
  "description": "Flooding near Main Street after heavy rain.",
  "category": "flood",
  "reported_at": "2025-03-12",
  "location": [
    {
        "longitude": 100.78,
        "latitude": 14.09
    }
   ],
  "media": [{"url": "photo1.jpg"}, {"url": "video1.mp4"}],
  "status": "resolved",
  "citizen": [
    {
        "first_name": "Anna",
        "last_name": "Smith",
        "thai_id": "7934763478"
    }
  ]
}
```

Document model diagram



The design of our system centers on grouping and status tracking. The Solutions document was created for grouping similar reported issues, meaning that one solution is usually used to fix several related issues at once. However, one issue can sometimes be so complex that it needs more than one specialized solution. That is why our reference relation is many to many.

The logic for tracking Issues and Solutions progress is as follows: the issue **status** depends entirely on the solution's date. The status is ‘pending’ when the issue has no Solutions reference. The status is 'in progress' if any linked solution is still active (no **finish_date**), and only becomes 'resolved' when all assigned solutions are complete. Finally, because one solution affects many people, the **Feedback** embedding is an array designed to collect multiple ratings and reviews from different affected citizens.

Identifying important insert, update, delete operations and transactions :

- Citizen report a new issue to government
- Citizen give more rating/feedback
- Government gives the solution for reported issues
- Deleted the incorrect issue

Identifying important data inquiries (a combination of simple and advanced useful business queries is preferable);

Queries from the last RDB project :

- Average time to fix issues (how much time did it take to close the solution) per office
- Remove resolved issues and solutions that are older than 2 years
- Update the current status of issue number 123 (transition ‘pending’→ ‘in progress’) and add a solution reference, plus add the issues_id in the solution table.
- Count the number of solutions that are completed and incompleted.
- Monthly statistics of closed solutions with average resolution time

New queries:

- Which office performs fixes the fastest on average?
- Which category has the most issues reported by citizens?
- Office with the highest and lowest solutions ratings.
- Total estimated costs for all solutions per month
- Identify 5 solutions that haven’t been finished
- List solutions completed this month
- Get the average solution feedback rating per office
- Find solutions without any feedback
- Find citizens who haven’t reported any issues but gave some feedback
- Total cost per office for this month

Queries:

Julianna:

- **Government gives solution for reported issues:**

```
db.solutions.insertOne({

  solution_id: 16,

  details: "Repair drainage system to prevent flooding in central area.",

  cost: 98000,

  issues: [{ issue_id: 4 }, { issue_id: 9 }],

  office: { name: "Public Works Department", contact: "02-123-4567" },

  start_date: ISODate("2025-03-01"),

  finish_date: null,
```

feedback: []

});

```
> db.solutions.insertOne({
    solution_id: 16,
    details: "Repair drainage system to prevent flooding in central area.",
    cost: 98000,
    issues: [{ issue_id: 4 }, { issue_id: 9 }],
    office: { name: "Public Works Department", contact: "02-123-4567" },
    start_date: ISODate("2025-03-01"),
    finish_date: null,
    feedback: []
  });
< {
    acknowledged: true,
    insertedId: ObjectId('68e879b2b1055d3089bbf75d')
  }
```

db.issues.updateMany(

{ issue_id: { $in: [4, 9] } },

{ $addToSet: { solutions: 16 }, $set: { status: "in progress" } }

);

```
> db.issues.updateMany(
    { issue_id: { $in: [4, 9] } },
    { $addToSet: { solutions: 16 }, $set: { status: "in progress" } }
  );
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 2,
    modifiedCount: 2,
    upsertedCount: 0
  }
```

- **Average time to fix issues (how much time did it take to close the solution) per office:**

db.solutions.aggregate([

{

$addFields: {

duration_days: {

```
            $cond: {
            if: { $and: [{ $ne: ["$finish_date", null] }, { $ne: ["$start_date", null] }] },
            then: { $divide: [{ $subtract: ["$finish_date", "$start_date"] }, 1000 * 60 * 60 * 24]
},
            else: null
            }
            }
            }
  },
  { $match: { duration_days: { $ne: null } } },
  {
            $group: {
            _id: "$office.name",
            avg_fix_time_days: { $avg: "$duration_days" },
            total_completed_solutions: { $sum: 1 }
            }
  },
  { $sort: { avg_fix_time_days: 1 } }
]);
```

```
{
  _id: 'City Electrical & Lighting Division (CEL)',
  avg_fix_time_days: 2.060185185185185,
  total_completed_solutions: 9
}
{
  _id: 'Traffic & Transport Planning (TTP)',
  avg_fix_time_days: 4.629166666666666,
  total_completed_solutions: 10
}
{
  _id: 'Municipal Law Enforcement (MLE)',
  avg_fix_time_days: 4.804487179487179,
  total_completed_solutions: 13
}
{
  _id: 'Environmental Services (ES)',
  avg_fix_time_days: 7.323717948717948,
  total_completed_solutions: 13
}
```

- **Monthly statistics of closed solutions with average resolution time:**
  db.solutions.aggregate:
  ([

```
{
    $match: { finish_date: { $ne: null } }
},
{
    $addFields: {
    duration_days: {
    $divide: [
    { $subtract: ["$finish_date", "$start_date"] },
    1000 * 60 * 60 * 24
    ]
    }
```

```
        }
  },
  {
        $group: {
        _id: {
        year: { $year: "$finish_date" },
        month: { $month: "$finish_date" }
        },
        closed_solutions: { $sum: 1 },
        avg_resolution_days: { $avg: "$duration_days" },
        total_cost: { $sum: "$cost" }
        }
  },
  { $sort: { "_id.year": 1, "_id.month": 1 } }
]);
```

```
< {
    _id: {
      year: 2023,
      month: 1
    },
    closed_solutions: 1,
    avg_resolution_days: 2.25,
    total_cost: 8500
  }
  {
    _id: {
      year: 2023,
      month: 2
    },
    closed_solutions: 1,
    avg_resolution_days: 3.625,
    total_cost: 15000
  }
  {
    _id: {
      year: 2023,
      month: 3
    },
    closed_solutions: 4,
    avg_resolution_days: 30,
    total_cost: 208000.5
  }
```

- **Total estimated costs for all solutions per month:**

```
db.solutions.aggregate([
 { $match: { finish_date: { $ne: null } } },
 {
        $group: {
        _id: {
        year: { $year: "$finish_date" },
        month: { $month: "$finish_date" }
```

```
        },
        total_cost: { $sum: "$cost" },
        total_solutions: { $sum: 1 }
        }
  },
  { $sort: { "_id.year": 1, "_id.month": 1 } }
]);
```

```
{
  _id: {
    year: 2024,
    month: 10
  },
  total_cost: 35000,
  total_solutions: 1
}
{
  _id: {
    year: 2025,
    month: 8
  },
  total_cost: 78500,
  total_solutions: 3
}
{
  _id: {
    year: 2025,
    month: 9
  },
  total_cost: 54000,
  total_solutions: 6
}
```

- **Get the average solution feedback rating per office:**

```
db.solutions.aggregate([
  { $unwind: "$feedback" },
  {
        $group: {
        _id: "$office.name",
        avg_rating: { $avg: "$feedback.rating" },
        total_feedbacks: { $sum: 1 }
        }
  },
  { $sort: { avg_rating: -1 } }
]);
```

```
{
  _id: 'City Electrical & Lighting Division (CEL)',
  avg_rating: 4.9,
  total_feedbacks: 10
}
{
  _id: 'Parks & Recreation Authority (PRA)',
  avg_rating: 4.8125,
  total_feedbacks: 16
}
{
  _id: 'Municipal Law Enforcement (MLE)',
  avg_rating: 4.785714285714286,
  total_feedbacks: 14
}
{
  _id: 'Environmental Services (ES)',
  avg_rating: 4.785714285714286,
  total_feedbacks: 14
}
```

Somkamon

- **Count the number of solutions that are completed and incompleted:**

```
db.issues.aggregate([
 {
  $group: {
   _id: {
    $cond: [
     { $eq: ["$status", "resolved"] },
     "resolved",
     "pending_or_in_progress"
    ]
   },
   count: { $sum: 1 }
  }
 }
])
```

```
{
  _id: 'resolved',
  count: 69
}
{
  _id: 'pending_or_in_progress',
  count: 31
}
```

- **Which category has the most issues reported by citizens**

```
db.issues.aggregate([
 {
  $group: {
   _id: "$category",
   total: { $sum: 1 }
  }
 },
 { $sort: { total: -1 }
 },
```

```
  {
    $limit: 1
  }
])
```

```
    _id: 'Road Infrastructure',
    total: 12
  }
```

- **List solutions completed this month**:

```
db.solutions.aggregate([
  {
    $addFields: {
      finishYear: { $year: { $toDate: "$finish_date" } },
      finishMonth: { $month: { $toDate: "$finish_date" } }
    }
  },
  {
    $match: {
      finishYear: new Date().getFullYear(),
      finishMonth: new Date().getMonth() + 1
    }
  },
  {
    $project: {
      _id: 0,
      solution_id: 1,
      finishYear: 1,
      finishMonth: 1
    }
  }
])
```

```
< {
    solution_id: 46,
    finishYear: 2025,
    finishMonth: 10
  }
  {
    solution_id: 48,
    finishYear: 2025,
    finishMonth: 10
  }
```

- **Total cost per office for this month:**

```
db.solutions.aggregate([
  {
   $addFields: {
    finishYear: { $year: { $toDate: "$finish_date" } },
    finishMonth: { $month: { $toDate: "$finish_date" } }
   }
  },
  {
   $match: {
    finishYear: new Date().getFullYear(),
    finishMonth: new Date().getMonth() + 1
   }
  },
  {
   $group:
   {
    _id: "$office.name",
    total_cost: {$sum: "$cost"},
    total_solution: {$sum: 1}
   }
  }
])
```

```
< {
    _id: 'Municipal Law Enforcement (MLE)',
    total_cost: 12000,
    total_solution: 1
  }
  {
    _id: 'Environmental Services (ES)',
    total_cost: 15000,
    total_solution: 1
  }
```

- **Citizen give more rating/feedback:**

```
db.solutions.updateOne(
  { solution_id: 3 },
  {
    $push: {
      feedback: {
        rating: 4,
        review: "The repair was good, but took a bit long.",
        citizen: {
          first_name: "Somjai",
          last_name: "Jairak",
          thai_id: "3948573928475"
        }
      }
    }
  })
```

```
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

```
{
  rating: 4,
  review: 'The repair was good, but took a bit long.',
  citizen: {
    first_name: 'Somjai',
    last_name: 'Jairak',
    thai_id: '3948573928475'
  }
```

Meta

- **Citizen report a new issue to government:**

```
db.issues.insertOne({
  issue_id: 2,
  description: "Streetlight not working on Soi 3, causing safety concerns at night.",
  category: "public lighting",
  reported_at: "2025-10-12",
  location: {
    district: "Rangsit",
    street: "Phahon Yothin Road, Soi 3"
  },
  media: [
    { type: "image", url: "streetlight_issue_soi3.jpg" }
  ],
  status: "new",
  citizen: {
    name: "Suthep Petchara",
    contact: "suthep.p@example.com"
  },
  solutions: []
});
```

```
>_MONGOSH
> db.issues.insertOne({
    issue_id: 101,
    description: "Streetlight not working on Soi 3, causing safety concerns at night.",
    category: "public lighting",
    reported_at: "2025-10-12",
    location: {
      district: "Rangsit",
      street: "Phahon Yothin Road, Soi 3"
    },
    media: [
      { type: "image", url: "streetlight_issue_soi3.jpg" }
    ],
    status: "new",
    citizen: {
      name: "Suthep Petchara",
      contact: "suthep.p@example.com"
    },
    solutions: []
  });
< {
    acknowledged: true,
    insertedId: ObjectId('68eb6d3b308a83ad934d814b')
```

- **Remove resolved issues and solutions that are older than 2 years**
  To find the cutoff date:

```
const cutoff = new Date();
cutoff.setFullYear(cutoff.getFullYear() - 2);
```

Then delete:

```
db.solutions.deleteMany({finish_date: { $lt: cutoff }});
```

```
> const cutoff = new Date();
  cutoff.setFullYear(cutoff.getFullYear() - 2);
< 1697102311425
> db.solutions.deleteMany({
    finish_date: { $lt: cutoff }
  });
< {
    acknowledged: true,
    deletedCount: 20
  }
```

- **Which office performs fixes the fastest on average?**

```
db.solutions.aggregate([  {
   $project: {
     office_name: "$office.name",
     duration_days: {
       $divide: [
         { $subtract: ["$finish_date", "$start_date"] },
         1000 * 60 * 60 * 24 ]}}},
 {
   $group: {
     _id: "$office_name",
     avg_fix_time_days: { $avg: "$duration_days" }
   } },
 {
   $sort: { avg_fix_time_days: 1 }  }]);
```

```
]);
< {
  _id: 'City Electrical & Lighting Division (CEL)',
  avg_fix_time_days: 2.060185185185185
}
{
  _id: 'Traffic & Transport Planning (TTP)',
  avg_fix_time_days: 4.629166666666666
}
{
  _id: 'Municipal Law Enforcement (MLE)',
  avg_fix_time_days: 4.804487179487179
}
{
  _id: 'Environmental Services (ES)',
  avg_fix_time_days: 7.323717948717948
}
{
  _id: 'Parks & Recreation Authority (PRA)',
  avg_fix_time_days: 33.68910256410256
}
{
```

- **Identify 5 solutions that haven't been finished:**

```
db.solutions.find( {
  $or: [
    { finish_date: { $exists: false } },
    { finish_date: null }
  ]},
 {
  _id: 1,
  solution_id: 1,
  "office.name": 1,
  start_date: 1,
  finish_date: 1
 }).limit(5);
```

```
< {
  _id: ObjectId('65e68372a5b81a2f98f4a22a'),
  solution_id: 36,
  office: {
    name: 'Public Works & Infrastructure Dept (PWI)'
  },
  start_date: 2025-01-20T08:00:00.000Z,
  finish_date: null
}
{
  _id: ObjectId('65e68372a5b81a2f98f4a251'),
  solution_id: 75,
  office: {
    name: 'Municipal Law Enforcement (MLE)'
  },
  start_date: 2027-08-01T09:00:00.000Z,
  finish_date: null
}
```

- **Find solutions without any feedback**

```
db.solutions.find(
 {
  $or: [
   { feedback: { $exists: false } }, { feedback: null },
   { feedback: {} }, ]  },
 {
  _id: 1,
  solution_id: 1,
  "office.name": 1,
  start_date: 1,
  finish_date: 1,
  feedback: 1
 });
```

```
< {
    _id: ObjectId('65e68372a5b81a2f98f4a22a'),
    solution_id: 36,
    office: {
      name: 'Public Works & Infrastructure Dept (PWI)'
    },
    start_date: 2025-01-20T08:00:00.000Z,
    finish_date: null,
    feedback: null
  }
  {
    _id: ObjectId('65e68372a5b81a2f98f4a251'),
    solution_id: 75,
    office: {
      name: 'Municipal Law Enforcement (MLE)'
    },
    start_date: 2027-08-01T09:00:00.000Z,
    finish_date: null,
    feedback: null
  }
```

Shamima

- **Deleted the incorrect issue**

```
db.issues.deleteOne({ issue_id: 123 });

db.issues.deleteOne({ description: "Duplicate test record" });

db.issues.deleteMany({ status: "incorrect" });
```

```
> db.issues.deleteOne({ description: "Duplicate test record" });
< {
    acknowledged: true,
    deletedCount: 0
  }
> db.issues.deleteOne({ issue_id: 123 });
< {
    acknowledged: true,
    deletedCount: 0
  }
> db.issues.deleteMany({ status: "incorrect" });
< {
    acknowledged: true,
    deletedCount: 0
  }
```

- **Update the current status of issue number 123 (transition ‘pending’→ ‘in progress’) and add a solution reference, plus add the issues_id in the solution table**

  db.issues.updateOne(
  { issue_id: 12},
  {
  $set: {
  status: "in progress",
  solutions: 7
  }
  }
  );

```
> db.issues.updateOne(
    { issue_id: 12},
    {
      $set: {
        status: "in progress",
        solutions: 7
      }
    }
  );
< {
    acknowledged: true,
    insertedId: null,
    matchedCount: 1,
    modifiedCount: 1,
    upsertedCount: 0
  }
```

- **Group issues by status and show issues' details**

db.issues.aggregate([
{
$group: {
_id: "$status",
issues: {
$push: {
issue_id: "$issue_id",
description: "$description",
category: "$category",
reported_at: "$reported_at",
location: "$location",
solution_id: "$solution_id",
citizen: "$citizen"
}
}
}
},
{
$project: {
_id: 0,
status: "$_id",
issues: 1

```
}
}
]);
```

```
    {
      issue_id: 89,
      description: 'Broken guardrail at the Lam Luk Ka canal crossing.',
      category: 'Public Safety',
      reported_at: 2025-10-11T02:00:00.000Z,
      location: {
        latitude: 13.96,
        longitude: 100.75,
        name: 'Lam Luk Ka, Canal Crossing'
      },
      citizen: {
        first_name: 'Supaporn',
        last_name: 'Jantima',
        thai_id: '96041000112'
      }
    },
    {
      issue_id: 91,
      description: 'Multiple lights out on the main highway billboard.',
      category: 'Safety and Lighting',
      reported_at: 2025-10-11T04:00:00.000Z,
      location: {
        latitude: 14.05,
        longitude: 100.61,
        name: 'Klong Luang, Highway 1'
      },
```

- **Find citizens who haven't reported any issues but gave some feedback**

```
db.solutions.aggregate([
  { $unwind: "$feedback" },
  {
    $lookup: {
      from: "issues",
      localField: "feedback.citizen.thai_id",
      foreignField: "citizen.thai_id",
      as: "reported_issues"
    }
  },
  { $match: { "reported_issues": { $size: 0 } } },
  {
    $project: {
      _id: 0,
      citizen: "$feedback.citizen",
      rating: "$feedback.rating",
      review: "$feedback.review"
    }
  }
```

])

```
  citizen: {
    first_name: 'Preeya',
    last_name: 'Surin',
    thai_id: '90070123445'
  },
  rating: 5,
  review: 'The area is completely clean, and the frequent trash collection schedule is working wonders.'
}
{
  citizen: {
    first_name: 'Preeya',
    last_name: 'Surin',
    thai_id: '90070123445'
  },
  rating: 5,
  review: 'The lot is clean now, and the surveillance is a great initiative.'
```

- **Office with the highest and lowest solution ratings**

```
db.solutions.aggregate([
{
$unwind: {
path: "$feedback",
preserveNullAndEmptyArrays: false
}
},
{
$group: {
_id: "$office.name",
avg_rating: { $avg: "$feedback.rating" },
total_solutions: { $sum: 1 },
contact: { $first: "$office.contact" }
}
},
{ $sort: { avg_rating: -1 } },
{
$group: {
_id: null,
highest_rated_office: { $first: "$$ROOT" },
lowest_rated_office: { $last: "$$ROOT" }
}
},
{
$project: {
_id: 0,
highest_rated_office: {
name: "$highest_rated_office._id",
```

```
contact: "$highest_rated_office.contact",
avg_rating: "$highest_rated_office.avg_rating",
total_solutions: "$highest_rated_office.total_solutions"
},
lowest_rated_office: {
name: "$lowest_rated_office._id",
contact: "$lowest_rated_office.contact",
avg_rating: "$lowest_rated_office.avg_rating",
total_solutions: "$lowest_rated_office.total_solutions"
}
}
}
])
```

```
{
  highest_rated_office: {
    name: 'City Electrical & Lighting Division (CEL)',
    contact: '02 987 65 43',
    avg_rating: 4.9,
    total_solutions: 10
  },
  lowest_rated_office: {
    name: 'Traffic & Transport Planning (TTP)',
    contact: '02 598 66 66',
    avg_rating: 4.6,
    total_solutions: 10
  }
}
```